

UAS Entrust nShield Connect HSM Deployment Guide - 6.0.1-GA (AI)

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1. Preface - UAS Entrust nShield Connect HSM Deployment Guide

This guide contains instructions on the installation, setup, and connection of Entrust nShield Connect to the client machine and the machine designated as the Remote File System (RFS) for the purpose of OneSpan (formerly VASCO) SEE.

Intended Reader

This document is intended as reference for administrators.

Document Scope

This guide covers the setup and configuration of the Entrust nShield Connect HSM.

You may refer to the Entrust nShield Connect documentation for installation steps and details regarding the configuration and setup of the HSM.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix UAS (Universal Authentication Server) documentation set:

- *AccessMatrix Release Notes*
- *[AccessMatrix Common Administration Guide](#)*
- *[UAS Administration Guide](#)*

2. Introduction to UAS Entrust nShield Connect HSM Deployment

The Entrust nShield Connect is a hardware security module (HSM) that provides secure cryptographic processing within a tamper-resistant casing. It is connected to a network by an Ethernet connection.

Remote Files System (RFS)

Each nShield Connect unit must have an RFS configured. You can configure the RFS on any computer, but it is normally located on the first client that is configured. The RFS contains master copies of all the files that the unit needs:

- **Module configuration file**
`%NFAST_KMDATA%\Key Management Data\<hsm-ESN>\config\`
- **Feature-enabling certificates**
`%NFAST_KMDATA%\Key Management Data\<hsm-ESN>\features\`
- **Encrypted Security World and key data for Security World**
`%NFAST_KMDATA%\Key Management Data\local\`



Note: Do not copy the module configuration file to file systems on other clients.

Client

- Each nShield Connect unit is configured to communicate with one or more client computers on the network.
- The client is the machine where Access Matrix is installed.
- Client servers can be privileged or unprivileged.
- To integrate UAS with Entrust HSM for Vasco SEE, the client MUST be configured as a privileged client.

Client	Rights
Privileged	○ Create security world ○ Load security world ○ Load SEE machine ○ Enable features
Unprivileged	○ Cannot run privileged commands ○ Use nShield Connect normally (load keys, use keys etc.)

Security World

- Entrust hardware security devices use the Security World paradigm to provide a secure environment for all your hardware security device and key management operations.
- The same security world is often loaded onto multiple HSM modules, this allows the modules to share keys with each other and the same group of card holders administer all the modules.
- An HSM module can only belong to one security world at any time.
- A Security World comprises:
 - an Administrator Card Set (ACS).
 - optionally, one or more Operator Card Sets (OCSs).
- The same security world can also be shared across multiple clients.

Smart Cards

The Security World uses:

- an Administrator Card Set (ACS) to control access to Security World configuration, as well as in recovery and replacement operations.
- one or more Operator Card Sets (OCSs) to control access to application keys.

Each card set consists of a total number of smart cards, **N**. The number of cards required to authorize an action is **K**. The required number K is known as the quorum. For example, the values for K/N being 2/3.

The **value for K should be less than N**. It is not recommended to create card sets in which K is equal to N, as an error on one card could render the whole card set unusable. If your ACS became unusable through such an error, you would have to replace the Security World and generate new keys.

3. Deploying Entrust HSM

The next sections provide information on deploying the Entrust nShield Connect HSM. Ensure that the following prerequisites are met before attempting the steps:

- Ensure that nShield Connect is safely and securely installed, and powers up successfully when you turn on the power supply at the rear of the module.
- Ensure that the client computer and the computer designated as your remote file system (RFS) are ready for NCipher Software installation.
- Ensure that both client and RFS machines were installed with JAVA 1.5x or latest version.
- Ensure that RFS and client machine's firewall are turned off.



Note: The RFS can be located on either the client or another network-accessible computer. If the RFS is installed on the client machine, the same file structure also contains the configuration files for that client. It is recommended to separate RFS & client into different machines for better job segregation.

3.1. Installing NCipher Security World Software

Software Versions

Software	Description
nCSS (nCIPHER Support Software)	• Basic software version, suitable for most deployments. • By default, the latest version of nCSS arrives with a new HSM. • This is almost certainly the version to deploy on production systems.
CipherTools	• Developer software version • Suitable for developing applications using commercial and nShield APIs • Contains necessary header files and documentation for development

 **Note:** Use nCSS, as it is the default software that comes together with new HSM. nCSS 11.70 is the latest version of nCSS.

Preparing Client Machine & RFS Machine

Refer to the following sections:

- Installing on Windows
- Installing on Linux

Installing on Windows

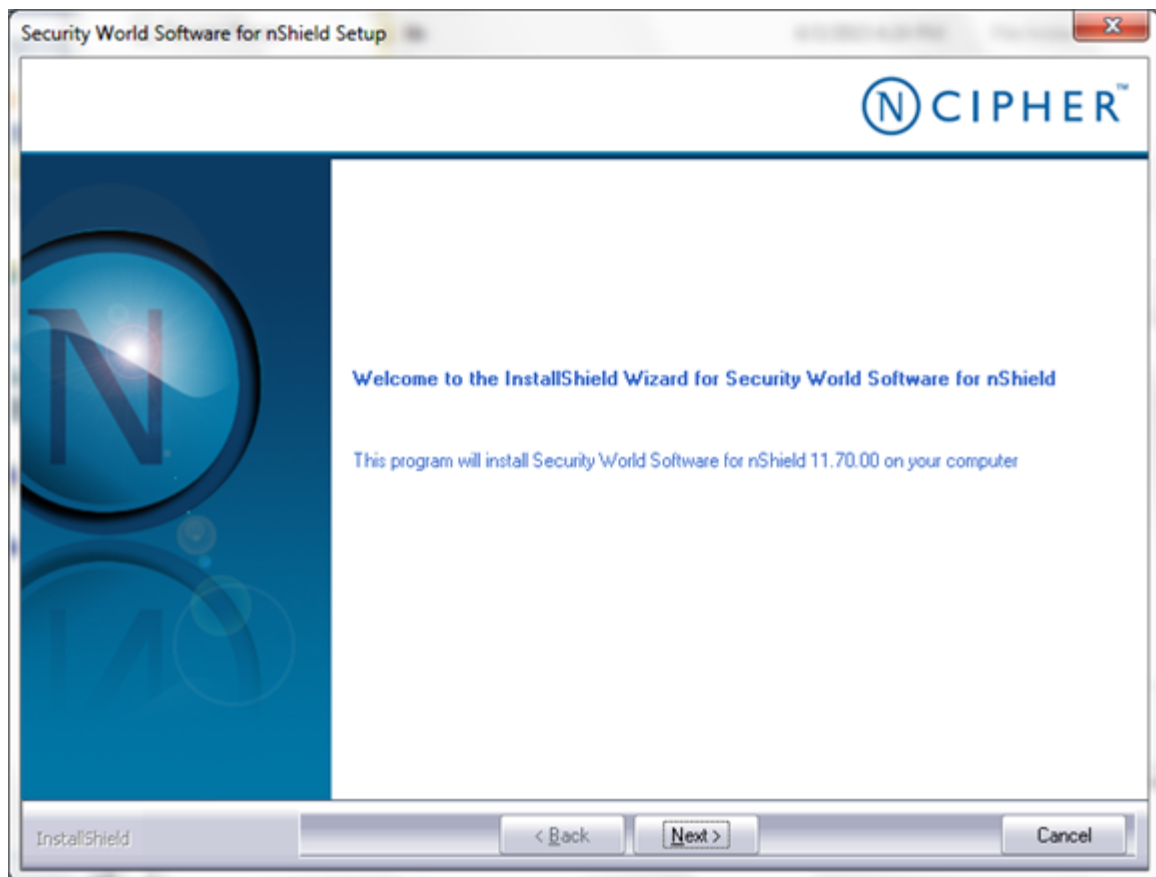
To install on Windows, perform the steps described below for both client and RFS machine:

1. Log in to Windows as Administrator.

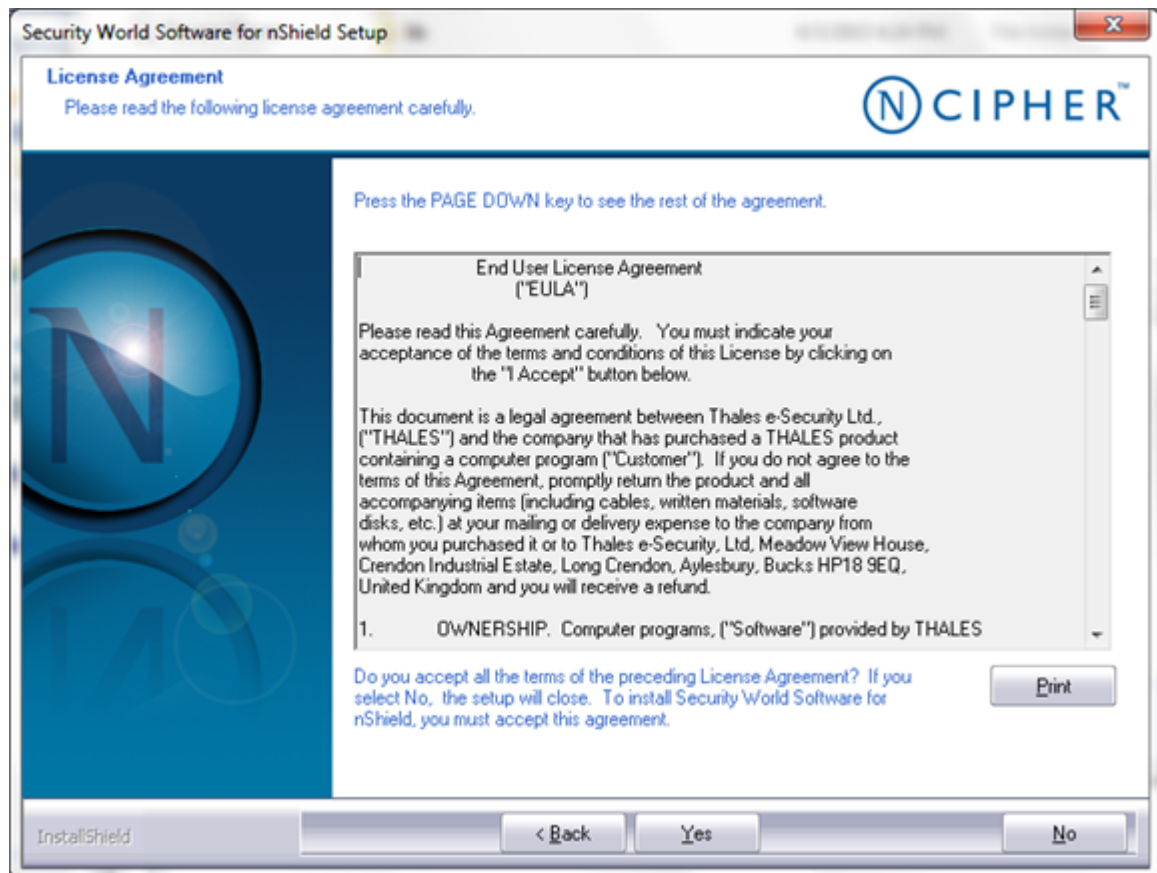
 **Note:** Administrator privilege is needed on each machine.

2. Extract out nShield Security World Software **SecWorld-win-user-11.70.00.iso** ISO image file then locate and run the **setup.exe** setup file (right click and run as administrator).

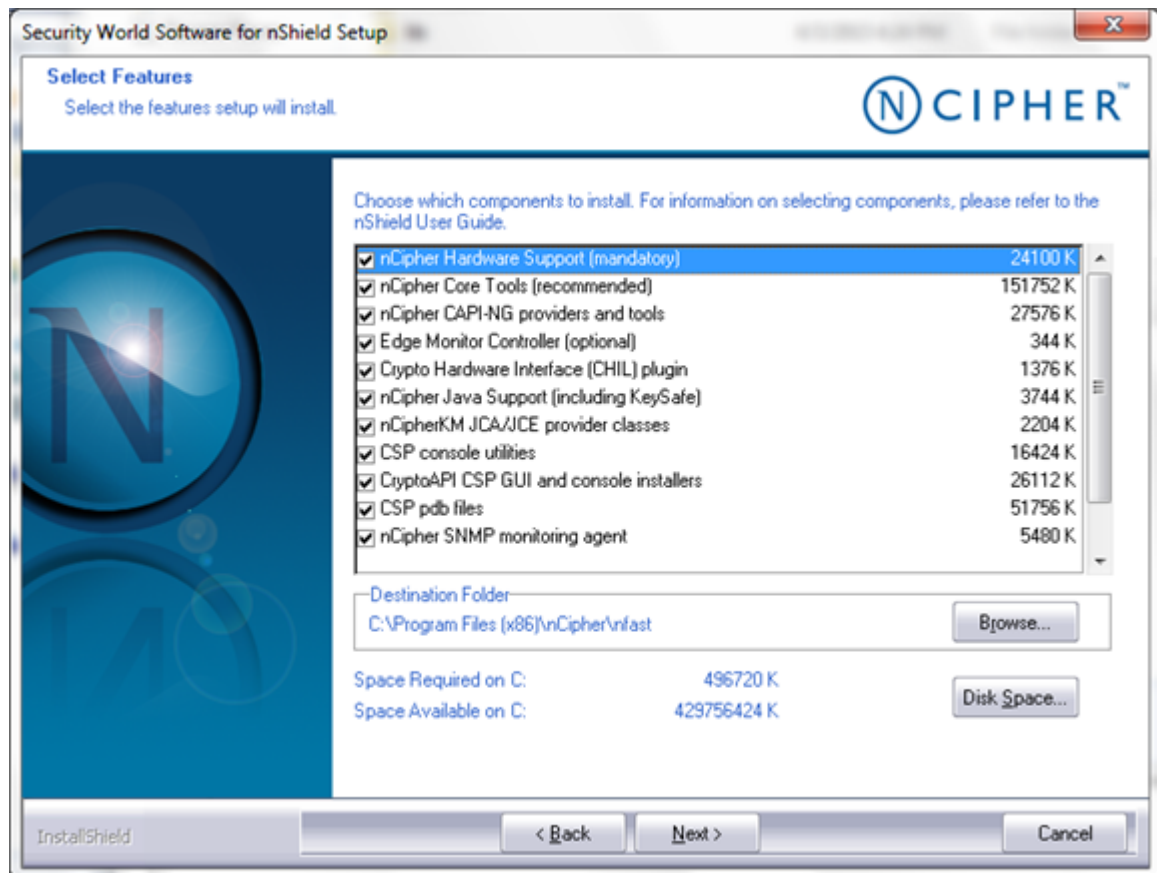
-
3. The following welcome screen is displayed. Click the **Next** button to continue.



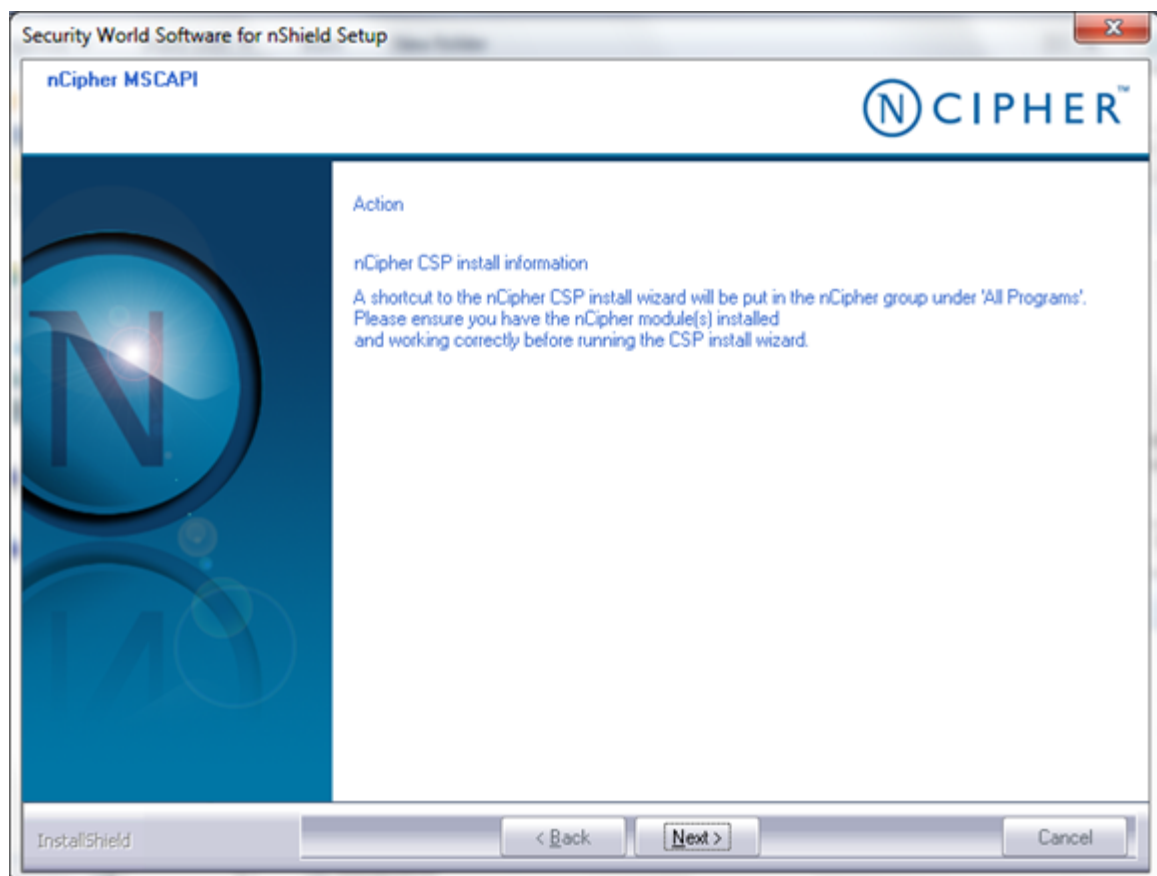
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4. Read the Software License Agreement and click the **Yes** button to accept and continue to the next screen.

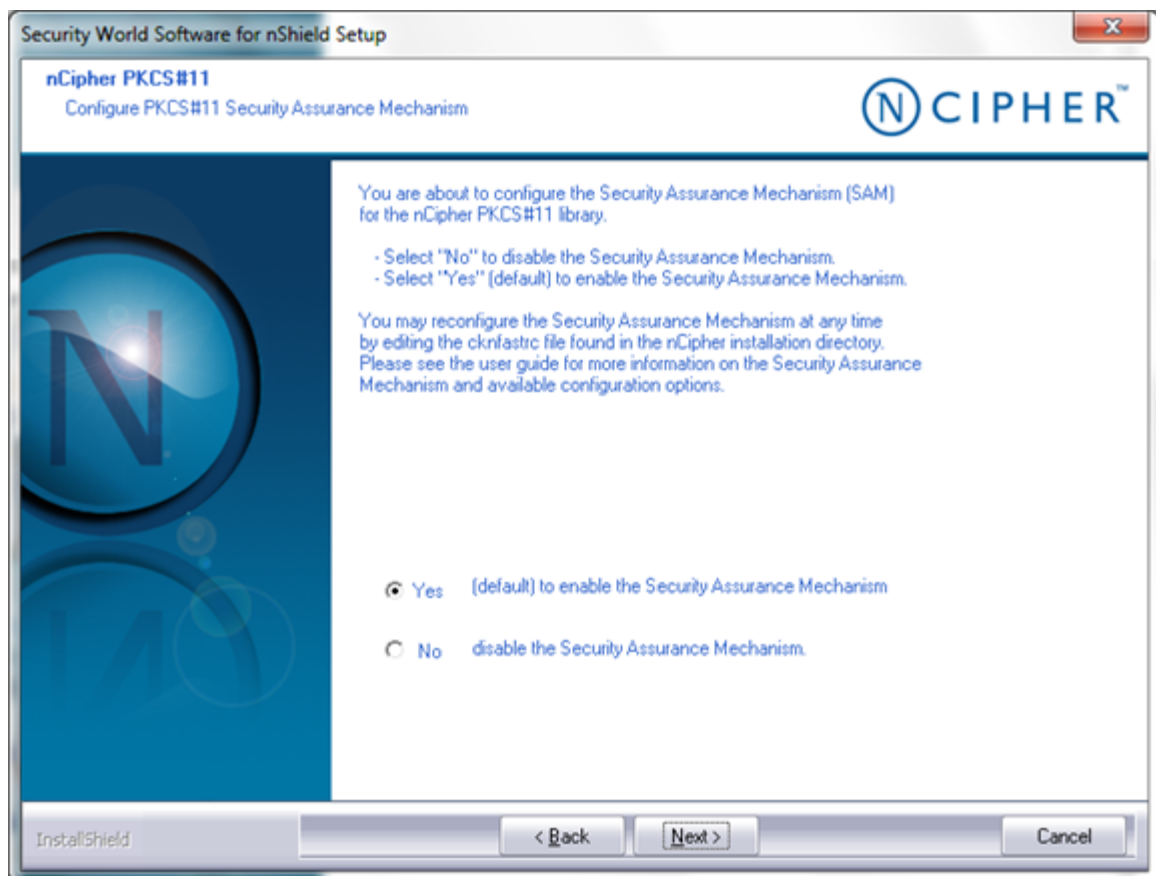
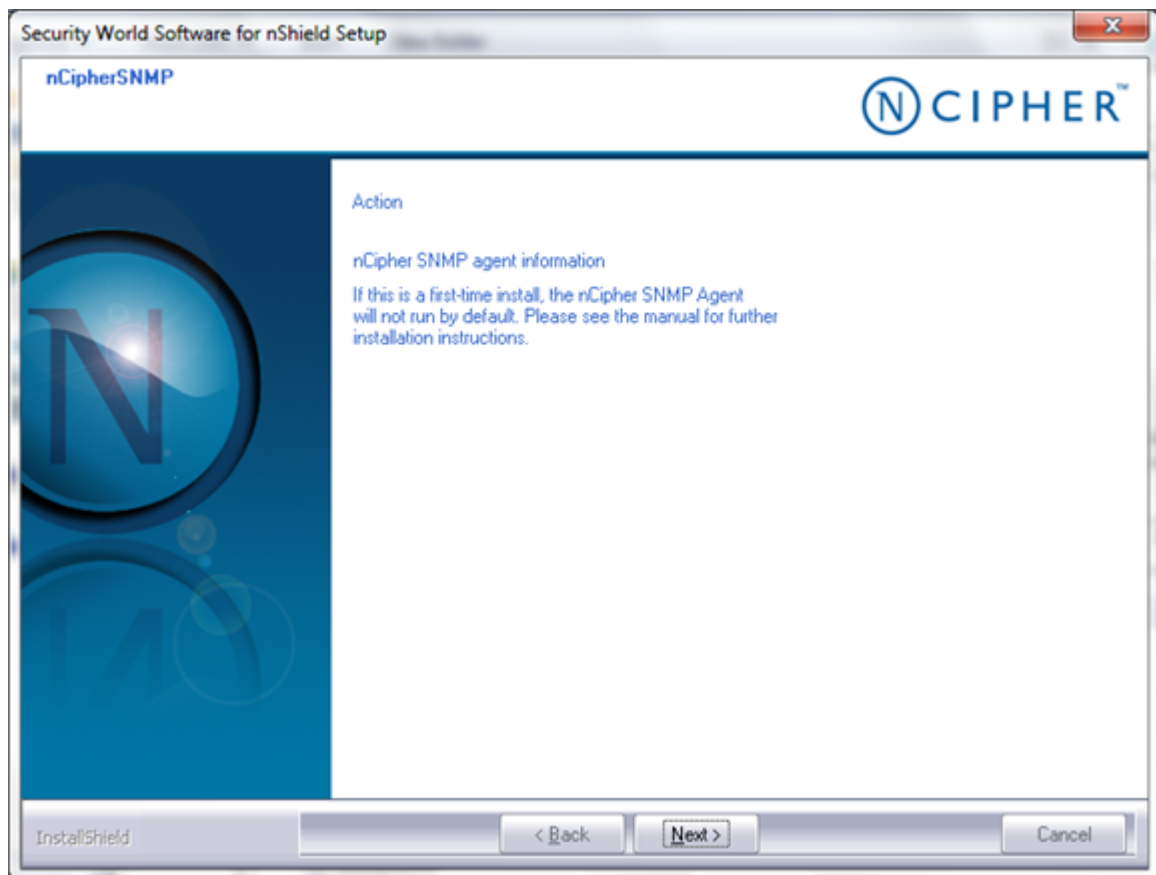


5. Select all features to install (default) and specify the installation folder and click the **Next** button to continue:

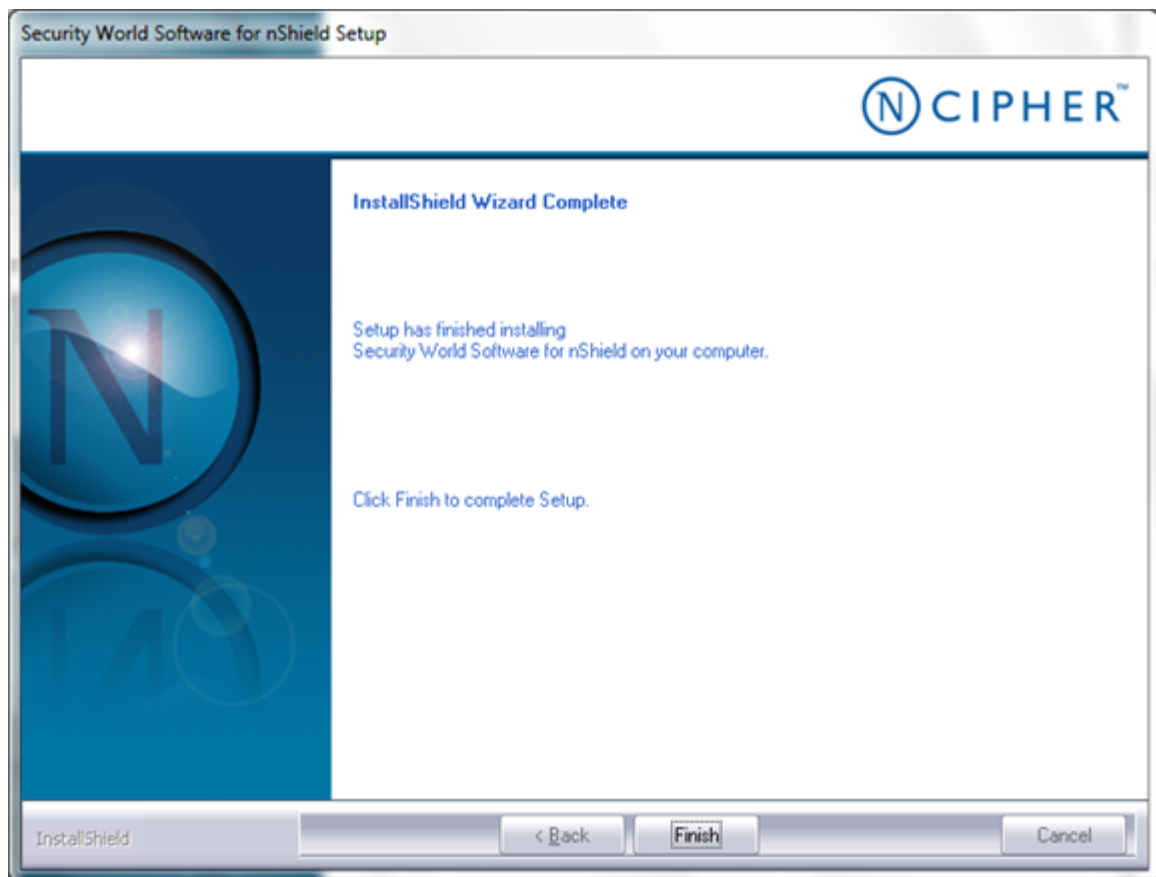


-
6. Click the **Next** button for the following screens:





7. Upon completion, click the **Finish** button to end the installation.



8. Once the installation is completed, the default files will be located in the following directories:

- %NFAST_HOME%:
 - (64 bit) *C:\Program Files (x86)\nCipher\infast*
 - (32 bit) *C:\Program Files\nCipher\infast*
- %NFAST_KMDATA%: *C:\ProgramData\nCipher*

File	Locations	Exist in RFS or Client
Security World files	<i>%NFAST_KMDATA%\Key Management Data\local\</i>	RSF, Client
Configuration file	<i>%NFAST_KMDATA%\Key Management Data\config\config\</i>	RSF, Client
Module configuration file	<i>%NFAST_KMDATA%\Key Management Data\<hsm-ESN>\config\</i>	RSF (generated during RFS setup)
Log files	Windows Event Viewer << nCipher Log >>	RSF, Client
Utilities	<i>%NFAST_HOME%\bin\</i>	RSF, Client

File	Locations	Exist in RFS or Client
Dynamic Feature Certificates	%NFAST_KMDATA%\Feature Certificates\	RSF, Client
Static Feature Certificates	%NFAST_KMDATA%\<hsm-ESN>\features\	RSF (generated during RFS setup)
Library files	%NFAST_HOME%\java and %NFAST_HOME%\toolkits\	RSF, Client

Installing on Linux

To install on Linux, perform the steps described below for both client and RFS machine:

1. Login to Linux as a user with root privilege.

Note: Root privilege is needed on each machine.

2. Open a terminal window, and change to the root directory.
3. Extract out NCipher Security World Software **SecWorld-linux64-user-11.70.00.iso** ISO image file.
4. Extract the required .tar files to install all the software bundles by running commands of the form:

```
tar xf <Extracted NCipher software path>/linux/<version>/nfast/<bundle>/<file>.tar
```

In this command, <version> is version of the operation system (e.g. libc_63), <bundle> is the directory name of a given bundle (e.g. hwsp or ctls), and <file>.tar is the name of a .tar file within a bundle directory.

Note: Some directories contain more than one .tar file.

5. Ensure the following component bundles and additional components are on the Security World:

Component Bundles	Additional Individual Components
○ hwsp (Hardware Support) - mandatory ○ ctls (Core Tools) - recommended ○ javasp (Java Support) - including KeySafe ○ mhsign (Signed netHSM firmware files)	○ hwcrhk (Crypto Hardware Interface (CHIL) plugin) ○ jcebsp (nCIPHERKM JCA/JCE provider classes) ○ ncsnmp (nCIPHER SNMP monitoring agent) ○ nhfw (nShield Connect and netHSM firmware) ○ pkcs11 (nCIPHER pkcs11 library)

-
6. Run the install script in the *hwsp* directory by using the following command:

```
/opt/nfast/sbin/install
```

7. The first time the installer runs (before any users or groups have been created), it displays a message about creating the *nfast* user:

```
User 'nfast' or group 'nfast' does not exist. To create the 'nfast' user under the
group 'nfast' in '/opt/nfast' home directory, select either one from the options
below:
a. Let this script run groupadd to create the group 'nfast'. Then, run useradd to add
the user 'nfast' to the group 'nfast' under '/opt/nfast' home directory. (Expected to
work with most Linux distributions).
b. Exit the script so you can create the user and group manually. Re-run the install
script when this is complete.
```

To enable the installer to create the user, group, and directory for you, type "1" to create the user, group and directory yourself; otherwise, type "2". Then press Enter.

8. Once installation is completed, the default files will be located in the following directories:
- `%NFAST_HOME%:/opt/nfast/`
 - `%NFAST_KMDATA%:/opt/nfast.kmdata`

File	Locations	Exist in RFS or Client
Security World files	<code>%NFAST_KMDATA%/local/</code>	RSF, Client
Configuration file	<code>%NFAST_KMDATA%/config/config/</code>	RSF, Client
Module configuration file	<code>%NFAST_KMDATA%/<hsm-ESN>/config/</code>	RSF (generated during RFS setup)
Log files	<code>%NFAST_HOME%/log/</code>	RSF, Client
Utilities	<code>%NFAST_HOME%/bin/</code>	RSF, Client
Dynamic Feature Certificates	<code>%NFAST_KMDATA%/femcerts/</code>	RSF, Client
Static Feature Certificates	<code>%NFAST_KMDATA%/<hsm-ESN>/features/</code>	RSF (generated during RFS setup)
Library files	<code>%NFAST_HOME%/java and %NFAST_HOME%/toolkits/</code>	RSF, Client

9. Login using a normal account.

10. Add `/opt/nfast/bin` to your PATH system variables:

- If you use the Bourne shell, add these lines to your system or personal profile:
-

```
PATH=/opt/nfast/bin:$PATH
export PATH
```

- If you use the C shell, add this line to your system or personal profile:

```
setenv PATH /opt/nfast/bin:$PATH
```

3.2. Setting Up nShield Connect Ethernet Interface

Configuring Module IP Address & Subnet Mask

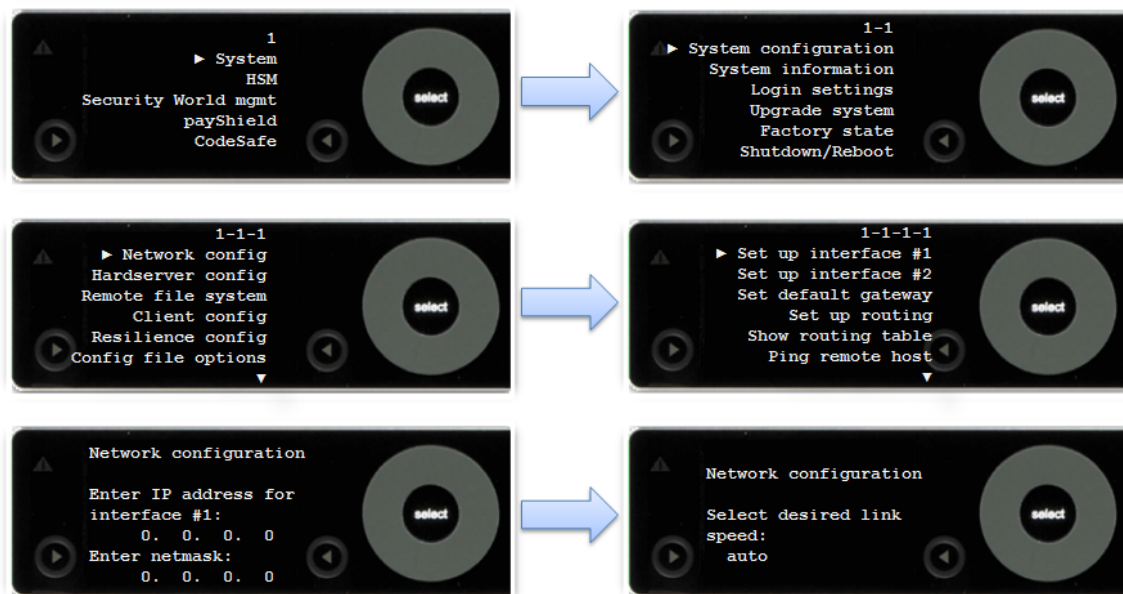
From the module's front panel:

1. Configure the Ethernet interface:
 - a. Enter the IP address of the module (from menu 1-1-1-1).
 - b. Enter the subnet mask.



Note: Do not use the same subnet if both interfaces are used.

- c. Leave link speed set to **auto / 1Gb**.
 - d. Select **Finish** and then reboot the module.
2. Refer to image below for the module front panel setup:



Configuring Gateway Connection

From the module's front panel:

1. Set the network location of the module by entering the default gateway (from menu 1-1-1-3).
 - a. Select **Finish** and **Continue**.
 2. To test on the connection, try to ping from the module to client and RFS machines (from menu 1-1-1-6).
-

-
3. Set the time and date on the module as UTC (from menu 1-1-8) and then reboot the module.
-

3.3. Setting Up RFS Machine

Configuring RFS to Use nShield Connect Module

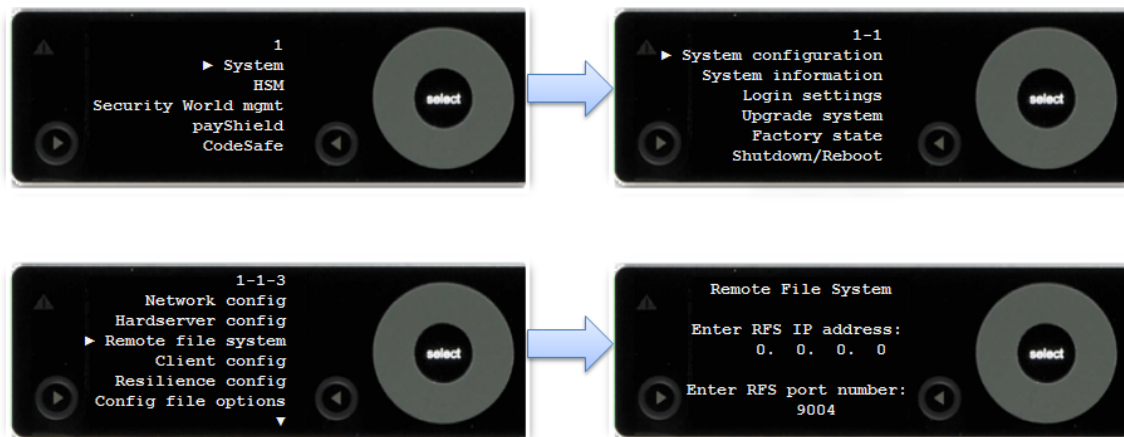
From RFS's command prompt:

1. Run command prompt as administrator in RFS machine and navigate to `%NFAST_HOME%\bin\`.
2. Retrieve the Eletronic Serial Number (ESN) and keyhash of the module by running the following command: `anonkneti <nShield Connect IP>` The ESN and keyhash are used in the command described in the next step.
3. Create the directory structure on the RFS by running the following command:
`rfs-setup --force <nShield Connect IP> <nShield Connect ESN> <nShield Connect KNETI HASH>`
4. After success, module config files on the RFS `%NFAST_KMDATA%\Key Management Data\<hsm-ESN>\config\` folder will appear.

From the module's front panel:

1. Configure the module to use the RFS (from menu 1-1-3) by entering the IP address of the RFS machine and leaving the port number as the default, "9004".
 2. Permit module config files on the RFS to be modified and then loaded to the module by turning on the auto push option (from menu 1-1-6-2):
 - a. Select **On**.
 - b. Enter the IP address of the RFS.
 - c. Select **Continue**.
-

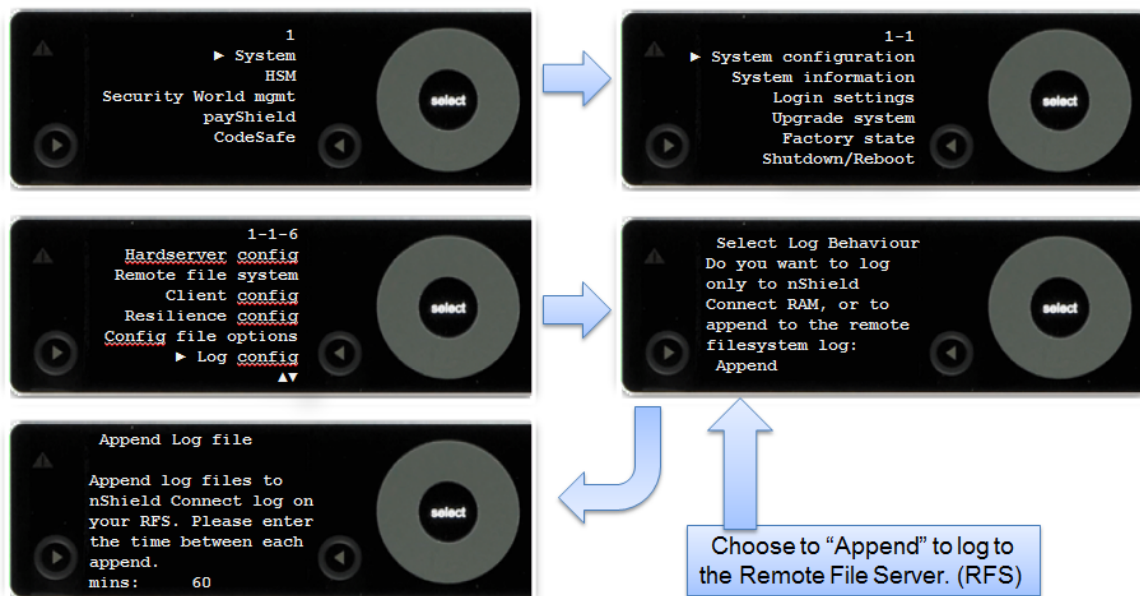
3. Refer to the image below for the module front panel setup:



Configuring Log to RFS

From the module's front panel:

1. Configure log file storage (from menu 1-1-6) by selecting one of the following options:
 - **Append:** stores the files on the module and RFS.
 - **Log:** stores the files on the module only.
2. Refer to the image below for the module front panel setup:



3.4. Setting Up Client Machine

Configuring Client to Use nShield Connect Module

i **Note:** (Optional) Install the nToken in the client machine. The client needs to use an nToken when its version is v11.50 or higher and it is communicating with a v11.40 or lower netHSM or nShield Connect.

From the module's front panel:

1. Configure the module to use the client machine:
 - a. Select **New client** (from menu 1-1-4-1).
 - b. Enter the remote client IP address.
 - c. If you want a privileged connection to the client, select **Priv.** on any port, else choose **Unprivileged**.
 - d. Select **No** to enroll the nToken and select the port (default "9004"). Make a note of the keyhash and the ESN or keep them on screen.

From client's command prompt:

1. Retrieve the ESN and keyhash of the module by running the following command:

```
anonkneti <nShield Connect IP>
```

The ESN and keyhash are used in the command described in the next step.

2. Run one of the following commands (refer to [Client](#) for privileged and unprivileged client details):

Privileged Client	nethsmenroll [Options] --privileged <nShield Connect IP><nShield Connect ESN><nShield Connect KNETI HASH>
Unprivileged Client	nethsmenroll [Options] <nShield Connect IP><nShield Connect ESN><nShield Connect KNETI HASH>

3. Configure the TCP sockets on the client for Java applications (for example, KeySafe) by running the command (only for privileged client):

```
config-serverstartup --enable-tcp --enable-privileged-tcpStop
```

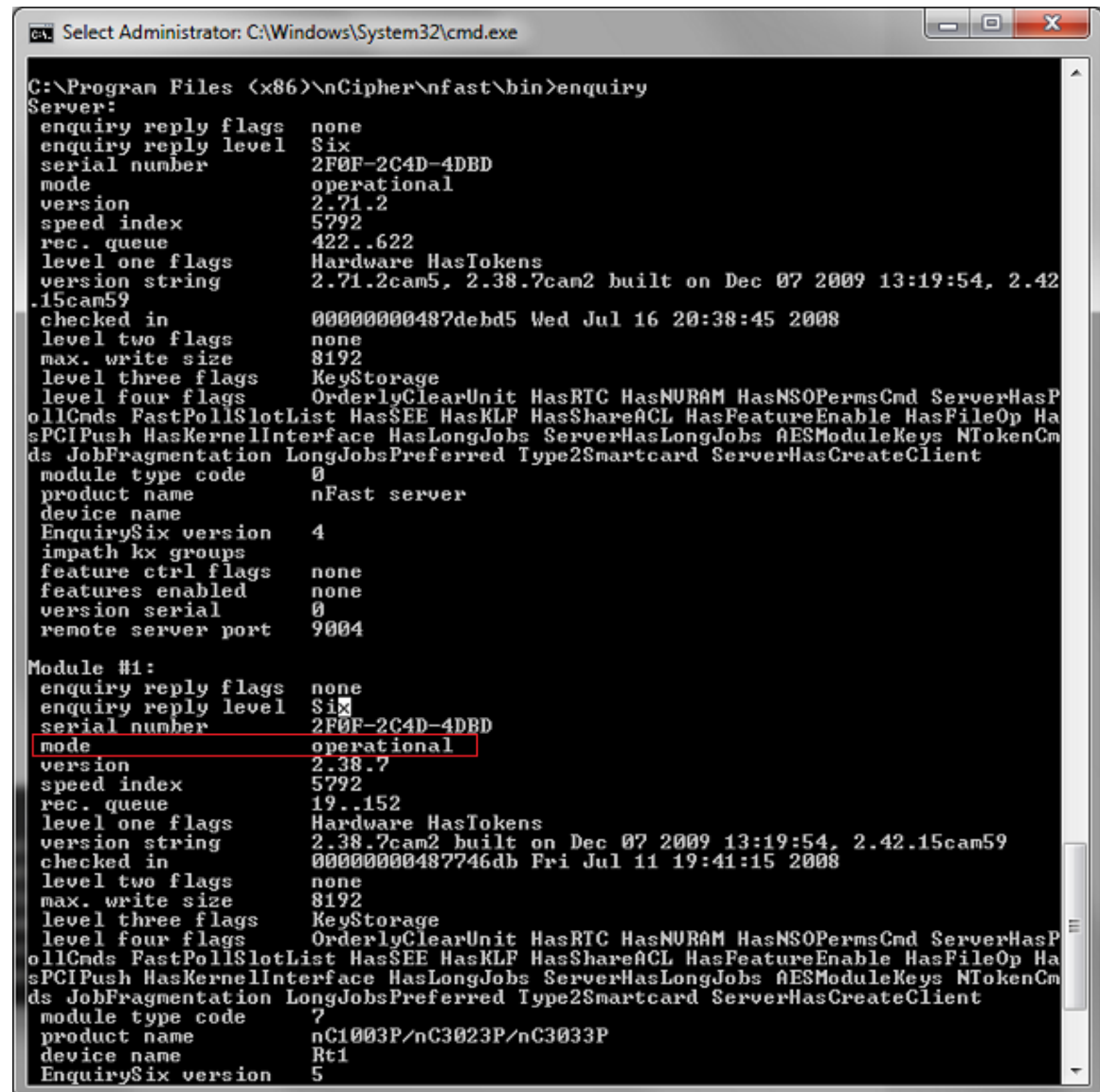
4. Restart the hardserver:

Windows	a. <code>net stop "nfast server"</code> b. <code>net start "nfast server"</code>
Linux	c. Log in as root and run the command: <code>/opt/nfast/sbin/init.d-ncipher restart</code>

5. Test the completed installation by running the command:

enquiry

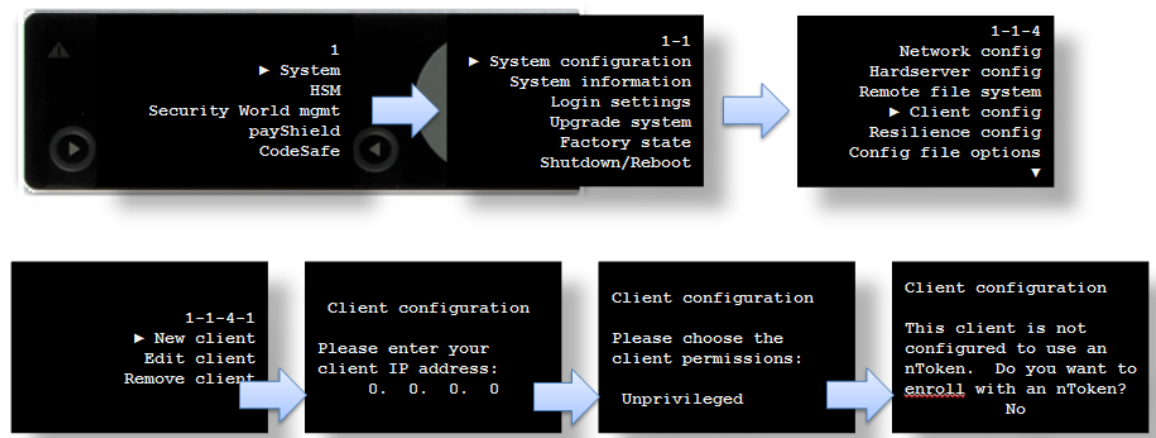
6. If the module is working correctly, the output from the enquiry utility should show mode operational in Module as shown below:



```
C:\Program Files (x86)\nCipher\nfast\bin>enquiry
Server:
enquiry reply flags    none
enquiry reply level    Six
serial number          2F0F-2C4D-4DBD
mode                   operational
version                2.71.2
speed index            5792
rec. queue             422.622
level one flags        Hardware HasTokens
version string         2.71.2cam5, 2.38.7cam2 built on Dec 07 2009 13:19:54, 2.42
.15cam59
checked in             00000000487debd5 Wed Jul 16 20:38:45 2008
level two flags        none
max. write size        8192
level three flags      KeyStorage
level four flags       OrderlyClearUnit HasRTC HasNURAM HasNSOPermsCmd ServerHasP
ollCnds FastPollSlotList HasSEE HasKLF HasShareACL HasFeatureEnable HasFileOp Ha
sPCIPush HasKernelInterface HasLongJobs ServerHasLongJobs AESModuleKeys NTokenCm
ds JobFragmentation LongJobsPreferred Type2Smartcard ServerHasCreateClient
module type code       0
product name           nFast server
device name
EnquirySix version     4
impath kx groups
feature ctrl flags     none
features enabled       none
version serial         0
remote server port     9004

Module #1:
enquiry reply flags    none
enquiry reply level    Six
serial number          2F0F-2C4D-4DBD
mode                   operational
version                2.38.7
speed index            5792
rec. queue             19.152
level one flags        Hardware HasTokens
version string         2.38.7cam2 built on Dec 07 2009 13:19:54, 2.42.15cam59
checked in             00000000487746db Fri Jul 11 19:41:15 2008
level two flags        none
max. write size        8192
level three flags      KeyStorage
level four flags       OrderlyClearUnit HasRTC HasNURAM HasNSOPermsCmd ServerHasP
ollCnds FastPollSlotList HasSEE HasKLF HasShareACL HasFeatureEnable HasFileOp Ha
sPCIPush HasKernelInterface HasLongJobs ServerHasLongJobs AESModuleKeys NTokenCm
ds JobFragmentation LongJobsPreferred Type2Smartcard ServerHasCreateClient
module type code       7
product name           nC1003P/nC3023P/nC3033P
device name            Rt1
EnquirySix version     5
```

7. Refer to image below for the module front panel setup:



3.5. Creating or Loading Security World

1. Before a Security World can be created via nShield Connect you must ensure that:
 - a. The module is configured correctly with an RFS.
 - b. The RFS must have no existing Security World files in its `kmdata/local/` folder.

From module's front panel:

1. Create either a New Security World or load an existing Load Security World:
 - a. **Create a new Security World** (from menu 3-2-1):
 - i. You need a blank set of Administrator Card Sets (ACS).
 - ii. Set an appropriate K-of-N **ACS quorum**.
 - iii. Choose to **Specify the quorum: Yes** or **No**. Choose **Yes** to enable/disable features such as key recovery and passphrase recovery. Choose **No** to accept defaults.
 - iv. If **Yes** was chosen, refer to the [Configure Quorum](#) section.
 - v. Create an **Operator Card Sets (OCS)** (from menu 3-5-1), choosing an appropriate K-of-N quorum and other options. A set of blank smart cards are required.
 - b. **Load an existing Security World** (from menu 3-2-2):
 - i. You need an existing Administrator Card Sets (ACS).
 - ii. Insert existing ACS card into the card slot at module front panel to finish loading.
 - iii. A set of OCS is ready to control access to application keys.



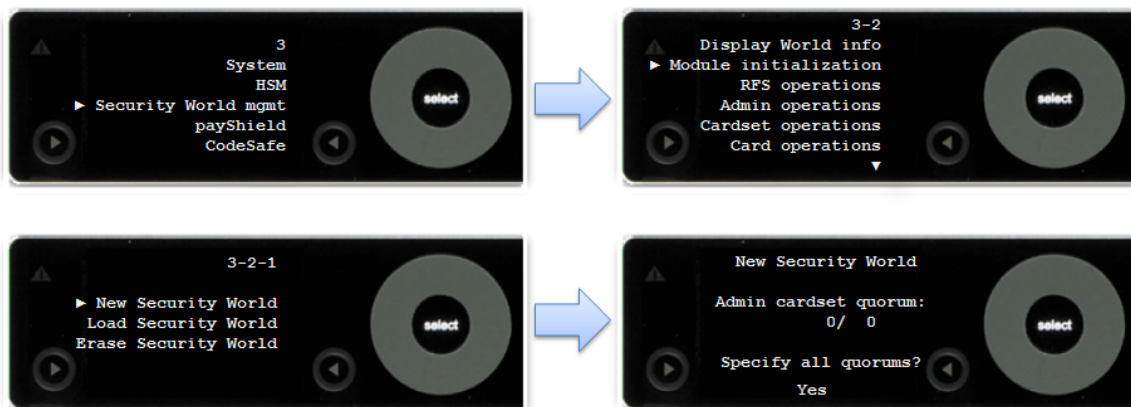
Note: Can create more OCS from menu 3-5-1.

From client's command prompt:

1. Run `nfkminfo` command to show the status of the Security World as shown in the image below:

```
C:\Program Files (x86)\nCipher\nfast\bin>nfkminfo
World
generation      2
state            0x17a70000 Initialised Usable Recovery PINRecovery !ExistingClient
RTC NURAM FTO   ?!AlwaysUseStrongPrimes SEEDebug
n_modules        1
hkns0            19c746e2c568d2816739a75d240eb0ae3d72e750
hkmm             d677dd1562dc382ab7ea38eea33b9b3d7238fa17 (type Rijndael)
hkmmwk           1d572201be533ebc89f30fdd8f3fac6ca3395bf0
hkre             da90f5b132c6dbbb5b82ddc65eaae6eca95f4414
hkra             7bb838126b917c1b18b3d2e4bb3b56369ab80b8a
hkmc             41d9674885bdaf0827ad3b797925605d9ecb6316
hkp              32ce0ae509e43a802186e18c4a11595c7138a673
hkrtc            a8e7e200af50ff91649d2c87ed200d3b556e933c
hknn             f10cf80eaad1a0d53ae0e9674984fa6497a8c2d7
hkdssee          72fe0823034c292b57be0b3d43893bad92f833f3
hkfto            c128ee27959c6d1a54e190942e01bfd46f4af68e
hknull           01000000000000000000000000000000000000000000
ex.client         none
k-out-of-n       1/2
other quora      m=1 r=1 p=1 nv=1 rtc=1 dsee=1 fto=1
createtime       2012-01-17 07:05:38
nso timeout      10 min
ciphersuite      DLf1024s160mRijndael
```

2. Please refer to image below for the module front panel setup steps. (Create New Security World)

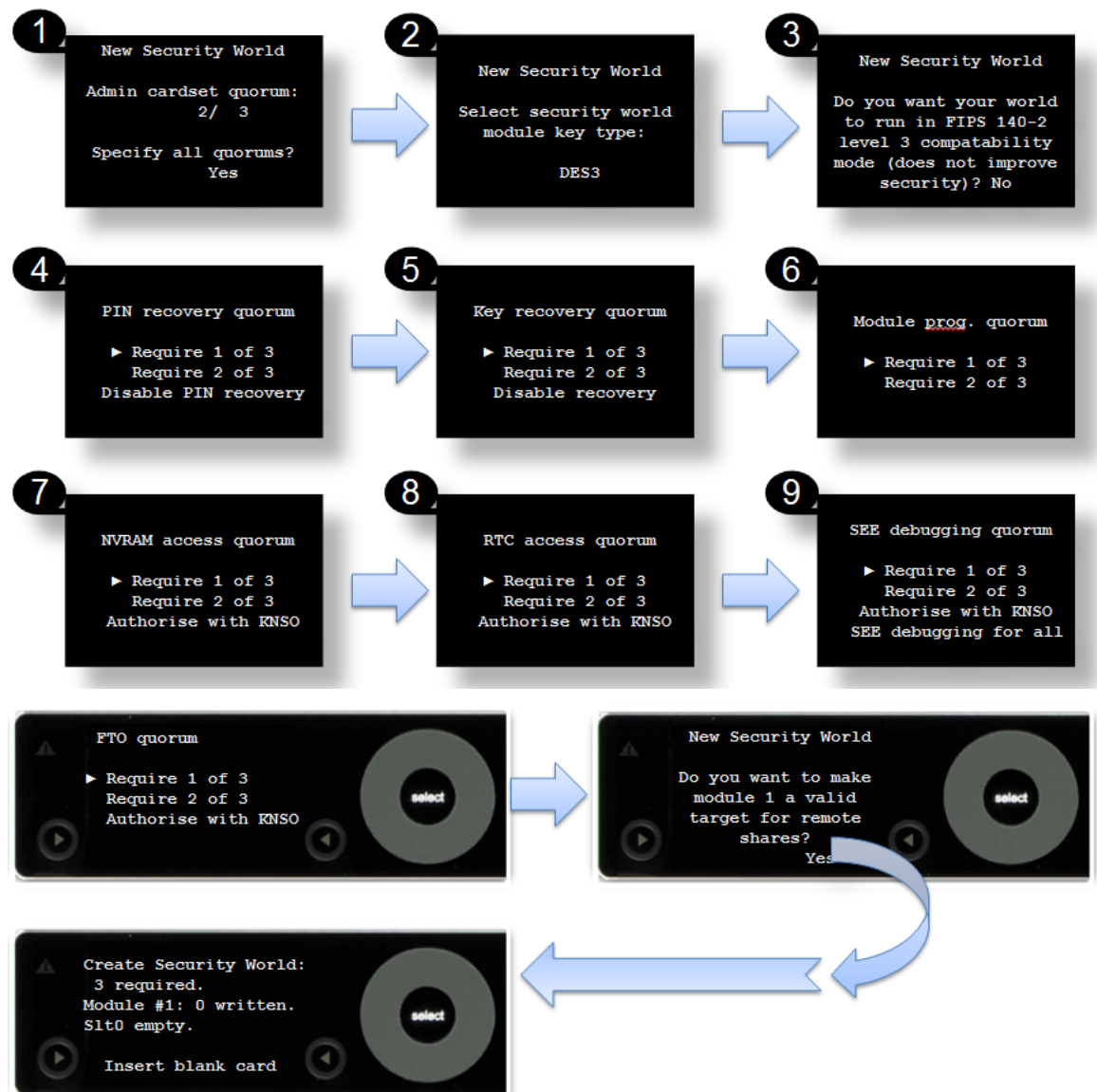


Configuring Quorum

From module's front panel:

1. If **Yes** is selected in **Specify all quorums**, select "DES3" for the **module key type**.
2. Choose **No** for security world to run in FIPS 140-2 level 3 compatibility mode.
3. Choose minimum require for each coming quorum.
4. Allow **module 1 a valid target for remote shares** by choosing **Yes**.
5. The remainder of screens prompt for **blank cards** and **passphrases** are optional.

6. Please refer to image below for the module front panel setup steps.



3.6. Synchronizing Client and RFS

The `rfs-sync` command-line utility is used to synchronize the `%NFAST_KMDATA%` directory between a cooperating client and the RFS it is configured to access.

Configuring Cooperation Client

From RFS's command prompt:

To allow other clients to access RFS, run the following command for each client:

With write access & keyhash authorization	(If connected to multiple modules, please run this for all modules) <code>rfs-setup --gang-client <client IP address> <nShield Connect ESN> <nShield Connect KNETI HASH></code>
With write access & without keyhash authorization	<code>rfs-setup --gang-client --write-noauth <client IP address></code>

i **Note:** The `--write-noauth` option should be used only if you believe your network is secure.

Configuring RFS-Sync

From client's command prompt:

On each client that is to be a cooperating client, run the following command:

With write access & keyhash authorization	<code>rfs-sync --setup <rfs IP address></code>
With write access & without keyhash authorization	<code>rfs-sync --setup --no-authenticate <rfs IP address></code>

Updating or Committing Configuration Files

Run `rfs-sync` whenever a cooperating client is initialized to retrieve data from the remote file system `%NFAST_KMDATA%/local/`, and also whenever a client needs to update its local copy of the data (or, if the client has write access, to commit changes to the data).

- All Client can pull files from the RFS to update by running command:
`rfs-sync --update`
 - Only privileged clients can push new (key/ cardset) files to the RFS by running command:
`rfs-sync --commit`
-

3.7. Setting Up KeySafe

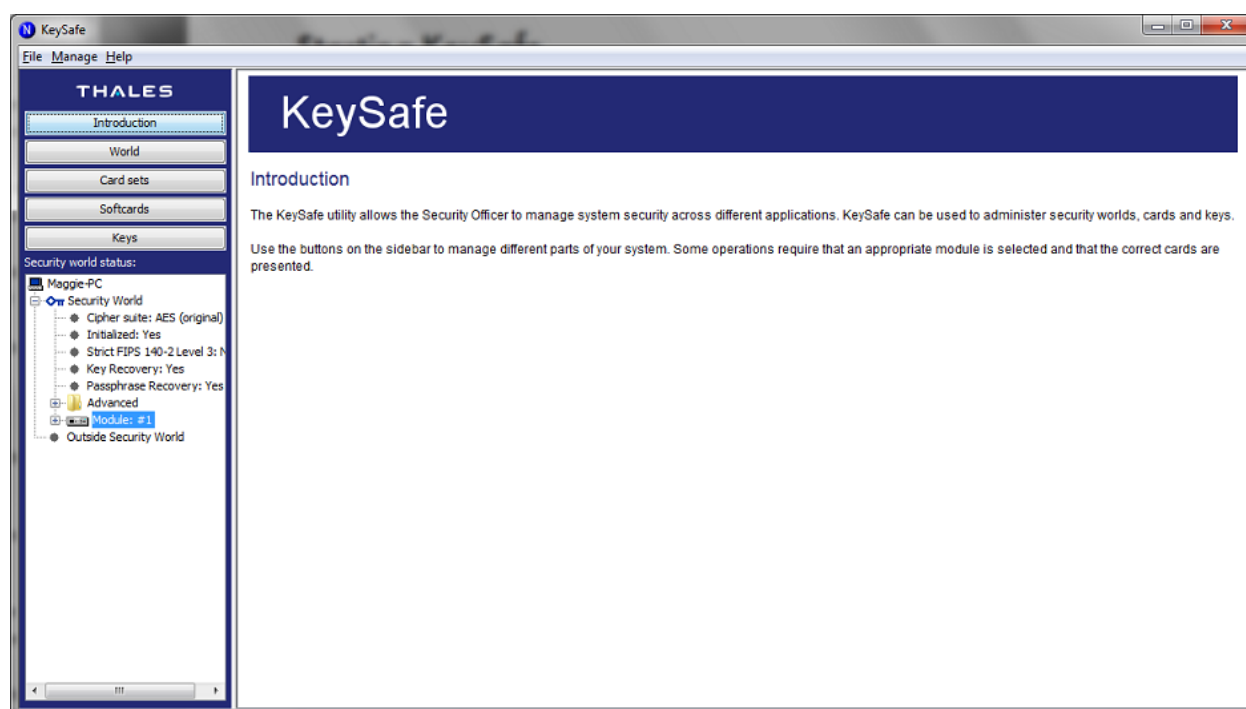
KeySafe provides a GUI-based interface to perform many of the main tasks required to use a Entrust security world:

- **Security World management**
 - **Card-set management**
 - **Key management**
1. Please ensure client with Java JRE/JDK 1.5, 1.6 or 1.7.
 2. The Security World Software must be installed.
 3. In the configuration file at `%NFAST_KMDATA%\config\config`, set the following port values in the server startup section:
 - a. `nonpriv_port=9000`
 - b. `priv_port=9001`
 4. Restart the server after this change.

Starting KeySafe

Windows	Start KeySafe by selecting the KeySafe entry from the Start menu under All Programs > nCipher.  Note: Administrator privileges is needed.
Linux	Start KeySafe by running the <code>/opt/nfast/bin/ksafe</code> script (assuming you installed the Security World Software in the default <code>/opt/nfast/</code> directory).

When it starts, KeySafe displays a window similar to image shown below:



KeySafe Functions

There are five menu buttons at the top of the sidebar:

Menu Buttons	Functions
Introduction	Clicking the Introduction menu button opens the introductory panel that KeySafe displays at startup.
World	Clicking the World menu button opens the World Operations panel, from which you can: • Initialize a Security World • Add modules to a Security World • Remove modules from a Security World Note: You cannot perform these operations on a module that is not in the pre-initialization mode.)
Card Sets	Clicking the Card Sets menu button opens the List Operator Card Sets panel, from which you can: • Examine or change an Operator Card Set or its pass phrase • Create a new Operator Card Set • Replace an Operator or Administrator Card Set • Discard an Operator Card Set

Menu Buttons	Functions
Softcards	Clicking the Softcards menu button opens the List Softcards panel, from which you can: • Create a new softcard • Change or recover the pass phrase on a softcard • Discard a softcard
Keys	Clicking the Keys menu button opens the Keys panel, from which you can: • Create a key • Import a key • Discard a key • View details of a key.
